ACADEMIUM ENGINE PACKET

Section 6: Response Assembly

Version: 2.0 **Status:** Active Specification **Scope:** Applies to every student-facing output across every turn in every case. Nothing reaches the student without passing through Response Assembly. **Runs after:** Section 3 (Consequence Generation), Section 4 (KPI Computation), Section 5 (Causal Explanation Generation) **Also receives:** PASS/NUDGE/BLOCK classification (engine constitution Section 6), session state, student original input **Outputs:** Complete structured response package delivered to the presentation layer. Also triggers session state update.

6.1 Purpose and Role

Response Assembly is the final step of the engine pipeline. It does not generate new content. Every word the student will read has already been produced by Sections 3, 4, and 5. Section 6 has three jobs:

Job 1 — Validate. Confirm that all upstream sections produced complete, internally consistent outputs. If anything is missing, contradictory, or malformed, Section 6 catches it here — not on the student's screen.

Job 2 — Assemble. Combine the validated outputs into a single structured response package, organized according to the rules for this specific turn type (First, Intermediate, or Final), input classification (PASS, NUDGE, or BLOCK), and case configuration.

Job 3 — Deliver. Pass the complete package to the presentation layer and trigger the session state update that enables compounding logic in subsequent turns.

The sequence is strict: Validate → Assemble → Deliver. No step may be skipped. If validation fails, the engine does not assemble a partial response and deliver it anyway. It resolves the failure first.

Why Assembly Is a Distinct Step

It would be simpler to deliver each section's output directly as it's generated. The reason this does not happen is consistency. A narrative that references stakeholder frustration must arrive alongside a KPI movement that reflects stakeholder dynamics. A causal explanation that references budget strain must not contradict a narrative that describes financial stability. The student's experience of the simulation depends on all three elements telling the same story. That coherence can only be enforced at a step that sees all three simultaneously — which is what Section 6 does.

Research Basis

Cognitive Load Theory (Sweller, 1988): Delivering narrative, metrics, and explanations as a coherent, intentionally structured package reduces extraneous cognitive load. The student does not have to reconcile contradictions between what the story says and what the numbers show. The assembled response is designed to be immediately readable and internally coherent.

Psychological Safety (Edmondson, 1999): A student who receives a contradictory or partially broken response — where the narrative says one thing and the metrics say another — will lose trust in the simulation. Response Assembly is the guardrail that prevents trust-breaking delivery failures.

Self-Determination Theory (Deci & Ryan, 1985): The response structure must consistently reinforce that the student's choices had meaningful, traceable effects. A well-assembled response — where narrative, metrics, and explanations all clearly connect back to the specific decision — reinforces the student's sense that their reasoning engaged with a real system.

6.2 Input Validation

Before any assembly begins, Section 6 performs a complete input validation check. All required inputs must be present, complete, and internally consistent.

Required Inputs Checklist

Input	Source	Required For
Consequence narrative	Section 3	All PASS turns
KPI movement set (direction, tier, KPI IDs, domain flags)	Section 4	All PASS turns
Causal explanation set (one per displayed KPI)	Section 5	All PASS turns
PASS/NUDGE/BLOCK classification	Engine Constitution Section 6	Every turn
Student's original input text	Input Reception	Every turn
Turn position (First/Intermediate/Final)	Session state	Every turn
RDS band	Section 2	All PASS turns
Session state (full prior history)	System	All turns

Input	Source	Required For
Next decision prompt	Case definition	Intermediate turns only
Final Outcome narrative	Case definition / generation	Final turn only
Reflection prompt	Case definition	Final turn only
Nudge content	Engine Constitution	NUDGE turns only
Redirect message	Engine Constitution	BLOCK turns only

Validation Failure Protocol

If any required input is missing:

1. **Step 1:** Section 6 logs the missing input and identifies which upstream section failed to produce it.
2. **Step 2:** Section 6 triggers a single retry of the failed section — passing the same inputs as the original call.
3. **Step 3:** If the retry produces valid output, assembly proceeds normally.
4. **Step 4:** If the retry fails, Section 6 applies the graceful degradation rules (Section 6.8) and delivers a partial response with an appropriate student-facing notification.

The retry is attempted once per missing input, not multiple times. The engine does not loop indefinitely. After one retry failure, it degrades gracefully.

6.3 Cross-Section Consistency Checks

Once all inputs are confirmed present, Section 6 performs three consistency checks before assembly. These are the engine's final quality control layer.

Check 1 — Narrative-KPI Domain Alignment

What it checks: Every KPI displayed to the student should correspond to an organizational domain that the consequence narrative references. If Team Morale is displayed as a movement but the narrative contains no reference to the team or internal organizational dynamics, the student will see a metric change with no story context.

How it works: Section 6 maps each displayed KPI to its domain (from Section 4.2) and scans the narrative for domain-relevant language. Exact phrasing is not required — the narrative must reference the general domain, not the specific metric.

Resolution when it fails: Section 6 flags the misaligned KPI. It checks whether the narrative can receive a brief domain reference without violating Section 3 quality gates. If yes, Section 6 triggers a targeted narrative extension (1 sentence maximum). If no, Section 6 removes the misaligned KPI from the display set and adjusts the explanation set accordingly.

Check 2 — Narrative-Explanation Contradiction

What it checks: No causal explanation should contradict a fact, stakeholder position, or organizational state established in the consequence narrative.

Examples of contradiction:

- Narrative: *"The client expressed confidence in the revised timeline."*
- K5 explanation: *"Stakeholder trust declined due to the client's uncertainty about the organization's capacity."* These cannot coexist. One must be wrong.

Resolution when it fails: The KPI movement direction or tier data from Section 4 is treated as the source of truth (it is the most constrained element, derived from pre-authored case architecture). Section 6 flags the contradiction and triggers a targeted regeneration of the conflicting narrative element only — not a full Section 3 regeneration.

Check 3 — Explanation-KPI Data Alignment

What it checks: Each causal explanation must correctly reference the direction and magnitude of the KPI it explains. An explanation for a Tier 2 positive movement must describe a moderate positive organizational effect — not a slight one, not a negative one.

Resolution when it fails: Section 6 triggers a targeted Section 5 regeneration for the misaligned explanation only, passing the correct KPI data. This is the least critical check — misalignment is usually a data reference error, not a substantive content problem, and targeted regeneration reliably resolves it.

6.4 Routing Logic: PASS, NUDGE, BLOCK

The PASS/NUDGE/BLOCK classification determines which assembly pathway is followed. These are three distinct response structures, not variations of the same structure.

PASS Pathway

Full pipeline output assembled. Complete response delivered. Session state updated to include this turn's data.

Triggers when: classification is PASS, OR when the student submits after 2 NUDGE cycles (treated as forced PASS per engine constitution).

NUDGE Pathway

No pipeline output assembled. The student's text is preserved in the input field. A nudge callout is generated and displayed adjacent to the input field — not in the main response area.

The nudge callout contains:

- 1–2 clarifying questions (pre-authored per turn or engine-generated within approved templates)
- No evaluation of the student's response
- No hint toward a specific answer

The nudge is non-blocking: the student may revise their response or submit as-is. If they submit as-is, the classification upgrades to PASS and the full pipeline runs on their original text.

Session state is NOT updated during a NUDGE cycle. The session records the nudge event but does not advance the turn counter.

BLOCK Pathway

No pipeline output assembled. The student's text is preserved in the input field (even if the text is hostile or profane — the student may wish to edit rather than retype). A redirect message is displayed from the safety posture templates (engine constitution Section 2.2).

The redirect message:

- Does not lecture the student
- Does not echo prohibited content back
- Redirects to the current decision prompt
- Is 1–2 sentences maximum

Session state is NOT updated. The turn counter does not advance. No retry limit applies — the engine redirects until the student provides a PASS-level response.

6.5 Complete Response Anatomy

Intermediate Turn Response (PASS)

This is the standard response structure for any turn between the First and Final turn.

```
ASSEMBLED RESPONSE — INTERMEDIATE TURN

[1] DECISION ACKNOWLEDGMENT (optional, 1 sentence)
    Purpose: Confirm what the engine understood the student to be doing.
    Rules:
    - MCQ turns: display the selected option text
    - FR turns: paraphrase the detected Intent signal only
    - If Intent signal was absent: omit the acknowledgment entirely
    - Never evaluate the decision
    - Never use second-person phrasing ("you decided")
    - Use organizational framing ("The direction taken: [paraphrase]")

[2] CONSEQUENCE NARRATIVE
    Source: Section 3 output (validated)
    Length: Per RDS band (80–160 words)
    Displayed as: Main body text in the response panel

[3] KPI MOVEMENT PANEL
    Source: Section 4 output (validated, max 3 KPIs)
    For each displayed KPI:
      - Contextual KPI label (per Section 4.9.1)
      - Direction indicator (↑ or ↓)
      - Magnitude label (Ligero / Moderado / Significativo)
      - "¿Por qué cambió esto?" / "Why did this change?" button
        → On activation: displays Section 5 causal explanation as overlay
        → Persistent: student can open/close at any time
        → Does not auto-display

[4] NEXT DECISION PROMPT
    Source: Case definition (pre-authored for this turn position)
    Displayed as: Separate panel below KPI movements
    Rules:
    - Prompt text is fixed (not generated dynamically)
    - For cases with trajectory variation: prompt selection is based on
      current session branch, determined by case architecture
    - Must be specific, actionable, scoped (per canonical case structure
      Section 4.3)
    - Input field is active and ready for student response
```

Final Turn Response (PASS)

```
ASSEMBLED RESPONSE — FINAL TURN
```

```
[1] DECISION ACKNOWLEDGMENT (optional, 1 sentence)
    Same rules as Intermediate turn

[2] CONSEQUENCE NARRATIVE
    Source: Section 3 output — Final turn variant
    No forward implication element (per Section 3.3 Element 4)

[3] KPI MOVEMENT PANEL
    Same structure as Intermediate turn
    Includes cumulative trajectory indicators (direction arrows showing
    overall session trend, not just this turn's movement)
    Source: Section 4 accumulation state

[4] FINAL OUTCOME NARRATIVE
    Source: Case definition / generation (Section 8 of canonical case structure)
    Length: 120–200 words
    Content requirements (from case structure):
    - Trajectory summary (narrative of the path taken — not a bullet list)
    - Sense of accomplishment (student navigated complexity; choices were
defensible)
    - Open-ended future (story is not fully resolved)
    - Never graded (no score, no grade language, no comparison to others)
    Displayed as: Distinct panel below KPI movements, clearly separated
    from the consequence narrative

[5] REFLECTION PROMPT
    Source: Case definition (pre-authored — one prompt per case)
    Display: Appears below Final Outcome narrative
    Rules:
    - Exactly one prompt
    - Skip button is always visible and accessible
    - No penalty for skipping
    - Input field active if student chooses to respond
    - Response is never evaluated, scored, or fed back into engine
    - Session marked complete after reflection submitted OR skipped
```

NUDGE Response

```
ASSEMBLED RESPONSE — NUDGE

[Student input field]
    Student's original text remains present and editable
```

```
    Cursor positioned at end of text (ready for editing)

[Nudge callout — displayed adjacent to input field]
    Not in the main response area
    Contains: 1–2 clarifying questions
    Visual treatment: distinct from the main response panel
    (callout box, tooltip, or inline message above input)
    Dismissible: yes — but input field remains active

[Submit button]
    Remains active — student can submit as-is or after editing
    Label: Same as standard submit (not "try again" — never implies failure)
```

BLOCK Response

```
ASSEMBLED RESPONSE — BLOCK

[Student input field]
    Student's original text preserved and editable
    Even if text is hostile or profane — never cleared automatically

[Redirect message — displayed above input field]
    1–2 sentences maximum
    Uses safety posture templates from engine constitution Section 2.2
    For profanity/hostility:
      "Entendemos tu frustración. Volvamos al caso — tu equipo necesita
       una decisión sobre [current situation]."
    For off-topic/nonsensical:
      "Este simulador está diseñado para trabajar con el caso actual.
       ¿Qué decisión tomarías respecto a [current decision prompt]?"

[Submit button]
    Active. No retry limit displayed. No counter shown to student.
```

6.6 Decision Acknowledgment Rules

The decision acknowledgment is the first element of an Intermediate or Final turn response. It is brief, optional in some conditions, and strictly constrained in what it can say.

When to Include It

Condition	Include Acknowledgment?
FR turn with STRONG Intent signal	Yes — paraphrase the intent
FR turn with PRESENT Intent signal	Yes — paraphrase the intent
FR turn with WEAK or ABSENT Intent signal	No — omit entirely
MCQ turn	Yes — display selected option text

What It Can Say

MCQ format: *"Opción seleccionada: [Option text as authored in the case]."*

FR format (when intent was detected): *"Dirección tomada: [1-sentence paraphrase of intent in organizational language]."*

What It Cannot Say

- Any evaluation of the decision: *"Good approach," "Interesting choice," "Bold direction"*
- Second-person phrasing: *"You decided to..."* → must be *"The direction taken..."*
- Any implication about what the decision means for the outcome: *"This will likely..."*
- Anything that doesn't directly reflect what the student's intent signal indicated

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6.7 Session State Update

After the validated response is delivered to the presentation layer, Section 6 triggers the session state update. This update is what enables compounding logic in all subsequent turns.

What Gets Written to Session State

```
SESSION STATE UPDATE (per turn)

turn_number: integer (1 through N, where N = professor-configured step count)
turn_position: "first" | "intermediate" | "final"
timestamp: ISO 8601
classification: "PASS" | "NUDGE" | "BLOCK"
student_input: string (original text, preserved exactly)
rds_band: "surface" | "engaged" | "integrated"
```

```
signals_detected: {
  intent: "strong" | "present" | "weak" | "absent",
  justification: "strong" | "present" | "weak" | "absent",
  tradeoff_awareness: "strong" | "present" | "weak" | "absent",
  stakeholder_awareness: "strong" | "present" | "weak" | "absent",
  ethical_awareness: "strong" | "present" | "weak" | "absent"
}
kpi_movements: [
  {
    kpi_id: "K1"–"K5",
    direction: "positive" | "negative",
    tier: 1 | 2 | 3,
    displayed: boolean
  }
]
competency_evidence: {
  C1: "demonstrated" | "emerging" | "not_evidenced",
  C2: "demonstrated" | "emerging" | "not_evidenced",
  C3: "demonstrated" | "emerging" | "not_evidenced",
  C4: "demonstrated" | "emerging" | "not_evidenced",
  C5: "demonstrated" | "emerging" | "not_evidenced"
}
narrative_delivered: string (consequence narrative text)
nudge_count: integer (0, 1, or 2 — for this decision point)
```

What Session State Enables

- **Section 3 compounding logic:** Full prior narrative history for stakeholder continuity and trajectory referencing
- **Section 4 accumulation model:** Prior KPI movement history for tier escalation, recovery attenuation, and trajectory classification
- **Section 5 signal anchoring:** Prior signal patterns available for reference across turns
- **Dashboard:** Competency evidence accumulates per student per session for reporting
- **Final Outcome narrative:** Full decision trajectory available for synthesis

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6.8 Graceful Degradation

When a pipeline step fails and retry does not resolve it, Section 6 applies degradation rules that preserve the student experience as much as possible while being transparent that something is not working as expected.

Degradation Hierarchy

Failed Section	What Student Receives	Student-Facing Message
Section 3 (narrative) fails	KPI movements + causal explanations only (no narrative)	*"El resumen narrativo no está disponible en este momento. Tus indicadores de impacto se muestran a continuación."*
Section 4 (KPIs) fails	Consequence narrative only (no metric panel)	*"Los indicadores de impacto no están disponibles en este momento. El análisis de tu decisión continúa a continuación."*
Section 5 (explanations) fails	Narrative + KPI movements; "Why?" button shows fallback	*"Esta explicación no está disponible en este momento."* (on "Why?" button activation)
Sections 3 + 4 fail	No content delivered	Full graceful error (see below)
All sections fail	No content delivered	Full graceful error

Full Graceful Error Message

Displayed when no pipeline output is available to assemble:

"Hubo un problema al procesar tu respuesta. Tu decisión y todo tu progreso han sido guardados. Por favor, intenta continuar en un momento. Si el problema persiste, contacta al soporte técnico."

["There was a problem processing your response. Your decision and all your progress have been saved. Please try to continue in a moment. If the problem persists, contact technical support."]

Graceful Degradation Rules

- **Student input is always preserved.** No degradation state ever clears or overwrites what the student submitted.
- **Session progress is always preserved.** Prior turns remain accessible in read-only mode regardless of the current turn's error state.
- **The turn counter does not advance during an error state.** The student retries the same turn when the issue is resolved.
- **Error messages never reveal technical details.** No stack traces, error codes, or system language visible to students.
- **Error messages never imply fault.** The message is neutral — *"there was a problem"* not *"your response caused an error."*

6.9 Quality Gates

All seven quality gates must pass before the response package is delivered. If any gate fails, the failure triggers the appropriate resolution protocol before delivery.

Gate	Check	Pass Condition	Resolution if Fail
G1: Input Completeness	Are all required inputs present for this turn type and classification?	All required inputs confirmed present	Retry failed section once; degrade if retry fails
G2: Narrative-KPI Domain Alignment	Does the narrative reference the domain of every displayed KPI?	Yes for all displayed KPIs	Targeted narrative extension (1 sentence) or KPI removal
G3: Narrative-Explanation Consistency	Do any causal explanations contradict the narrative?	Zero contradictions	Targeted narrative regeneration (KPI data is source of truth)
G4: Explanation-KPI Data Alignment	Does each explanation correctly reflect the direction and tier of its KPI?	All aligned	Targeted explanation regeneration (Section 5)
G5: Routing Accuracy	Is the correct assembly pathway applied (PASS/NUDGE/BLOCK, First/Intermediate/Final)?	Correct pathway confirmed	Re-route to correct pathway
G6: Student Text Preservation	For NUDGE and BLOCK: is the student's original text intact and present in the input field?	Yes — unchanged	Restore from input reception record
G7: No New Prohibited Content	Does the assembled package introduce any prohibited language not caught in upstream sections?	Zero prohibited terms in full assembled package	Identify and remove; regenerate if content is load-bearing

6.10 Worked Examples

EXAMPLE A — Intermediate Turn, Engaged Band, PASS

Turn position: Turn 4 of 8 **RDS band:** Engaged (RDS = 7) **Input classification:** PASS **Intent signal:** PRESENT ("Priorizaría la comunicación con el equipo") **KPIs to display:** K2 ↑ Tier 2, K4 ↓ Tier 1

Assembly sequence:

```
VALIDATION
✓ Narrative present (128 words, Engaged band)
✓ KPI set present: K2 ↑ T2, K4 ↓ T1
✓ Explanations present: K2 explanation (3 sentences), K4 explanation (2 sentences)
✓ Next decision prompt confirmed for Turn 5

CROSS-SECTION CHECKS
✓ Check 1: Narrative references "equipo" (K2 domain) and "entregas" (K4 domain)
✓ Check 2: No narrative-explanation contradictions found
✓ Check 3: Explanations correctly reference Tier 2 positive / Tier 1 negative

ROUTING: PASS → Intermediate Turn pathway

ASSEMBLED RESPONSE:

[1] DECISION ACKNOWLEDGMENT
"Dirección tomada: priorizar la comunicación interna del equipo antes de asumir
compromisos externos."

[2] CONSEQUENCE NARRATIVE
[128-word narrative from Section 3 — Engaged band, Turn 4 with compounding from
prior turns]

[3] KPI PANEL
↑ Cohesión del Equipo    [Moderado]   [¿Por qué cambió esto? →]
↓ Capacidad de Entrega   [Ligero]     [¿Por qué cambió esto? →]

[4] NEXT DECISION PROMPT
[Pre-authored Turn 5 prompt from case definition]
[Active input field]

SESSION STATE UPDATE: written ✓
```

EXAMPLE B — Final Turn, Integrated Band, PASS

Turn position: Turn 8 of 8 (Final) **RDS band:** Integrated (RDS = 13) **KPIs displayed:** K2 ↑ Tier 1, K5 ↑ Tier 2, K4 ↓ Tier 1

Assembly sequence:

```
VALIDATION
✓ Narrative present (158 words, Integrated band, no forward implication)
✓ KPI set present with cumulative trajectory data
✓ Explanations present for all three KPIs
✓ Final Outcome narrative present (165 words)
✓ Reflection prompt present (pre-authored)

CROSS-SECTION CHECKS
✓ All three domain checks pass
✓ No contradictions
✓ Explanation-KPI alignment confirmed

ROUTING: PASS → Final Turn pathway

ASSEMBLED RESPONSE:

[1] DECISION ACKNOWLEDGMENT
"Dirección tomada: consolidar la relación con las partes clave a través de
compromisos transparentes sobre capacidad real."

[2] CONSEQUENCE NARRATIVE
[158-word narrative — Final turn, full compounding across 8 turns visible]

[3] KPI PANEL (with cumulative trajectory indicators)
↑ Cohesión del Equipo        [Ligero]      [¿Por qué cambió esto? →]
   Trayectoria de sesión: ↑ positiva
↑ Confianza de Partes Clave  [Moderado]    [¿Por qué cambió esto? →]
   Trayectoria de sesión: ↑ mixta
↓ Capacidad de Entrega       [Ligero]      [¿Por qué cambió esto? →]
   Trayectoria de sesión: ↓ negativa

[4] FINAL OUTCOME NARRATIVE
[165-word trajectory summary from case definition/generation —
trajectory-informed, never graded, open-ended future]

[5] REFLECTION PROMPT
"¿Cuál fue la decisión más difícil que tomaste y qué te llevó a tomarla?"
[Optional input field]            [Omitir / Skip]

SESSION STATE UPDATE: written ✓
SESSION STATUS: marked complete (pending reflection submission or skip)
```

EXAMPLE C — NUDGE Assembly

Classification: NUDGE (first nudge — student wrote "Es una situación difícil") **No pipeline sections run**

```
ASSEMBLY:

[Student input field — original text preserved]
"Es una situación difícil"
[Cursor at end — ready to edit]

[Nudge callout — adjacent to input field]
"Tu observación es válida. Para poder avanzar, considera:
 ¿Qué acción concreta tomarías en esta situación y por qué?
 ¿Qué estarías priorizando sobre otras alternativas?"

[Submit button — active, standard label]

SESSION STATE: nudge_count updated to 1. Turn not advanced.
```

EXAMPLE D — BLOCK Assembly

Classification: BLOCK (empty input submitted)

```
ASSEMBLY:

[Student input field — empty, or whatever the student submitted]

[Redirect message — above input field]
"Este simulador está diseñado para trabajar con el caso actual.
¿Qué decisión tomarías respecto a la situación del equipo de operaciones?"

[Submit button — active]

SESSION STATE: BLOCK event logged. Turn not advanced. No counter shown.
```

6.11 Section Summary: What the Engine Must Do

At every turn, the response assembler must:

1. **Validate all required inputs** before assembling. Never deliver a response built on incomplete or unvalidated data.
2. **Run all three cross-section consistency checks** — narrative-KPI alignment, narrative-explanation consistency, explanation-KPI data alignment. Resolve all failures before assembly proceeds.
3. **Route to the correct pathway** — PASS/NUDGE/BLOCK, combined with First/Intermediate/Final turn type. These are distinct structures, not variations.
4. **Assemble the complete response package** — all required elements for the pathway, in the correct order, with all conditional logic applied (acknowledgment rules, cumulative trajectory indicators on Final turn, etc.).
5. **Apply graceful degradation** if any upstream section failed and retry did not resolve it. Never deliver a partial response without a student-facing notification.
6. **Pass all seven quality gates** before delivery. Gate failures trigger targeted resolution, not full pipeline restart.
7. **Deliver to the presentation layer.** The engine produces a structured package; the UI renders it. The engine does not control visual layout.
8. **Trigger the session state update** after delivery — writing all turn data required for compounding logic, dashboard reporting, and Final Outcome synthesis.

Section 6 is the terminal section of the engine pipeline. It connects to all upstream sections (Sections 2–5) as a receiver and consumer of their outputs. Its downstream connections are: the Presentation Layer (receives the assembled response package) and the Session State (receives the turn data update that enables all subsequent turns to compound correctly).